

AIF 2026 Project Ideas Catalog

A Comprehensive Selection Guide for AI Fellows (Research & Development Tracks)

CRITICAL CRITERION: Projects must not be simple RAG models or standard tools whose execution can be fully automated out-of-the-box by general LLMs like Claude. Every proposal demands advanced implementations, such as multi-agent collaboration frameworks, localized speech/vision workflows, or edge-optimized architecture.

1. HEALTH & MEDICAL TECHNOLOGY

Implantable Device Identification in X-rays

Supervisor: [Rakshya Lama Moktan](#) • Type: [Vision, Prototyping](#)

PROBLEM STATEMENT

Missing documentation and records require manual X-ray checks for cardiac devices. Accidental or unverified exposure to MRI fields can lead to catastrophic failures and severe patient risks.

KEY GOALS

- Automated bounding box localization of cardiac devices on standard chest radiographs.
- Fine-grained classification models to identify exact manufacturers and device models.
- Multi-task analysis to instantly generate specific MRI safety and compatibility profiles.

APPLICATION AREAS

Radiology & Imaging, Emergency Medicine, Clinical Decision Support Systems, Hospital Administration.

PROJECT SUMMARY

A computer vision framework that localizes and identifies cardiac devices within chest scans to automatically flag manufacturer data, model parameters, and strict MRI safety restrictions.

CBT & DBT Companion

Supervisor: [Raj Kumar Biswokarma](#) • Client: [Individual/Person](#) • Type: [Vision, Prototyping](#)

PROBLEM STATEMENT

Real-time mental health coaching is highly limited by the availability and cost of specialists. Patients frequently struggle to regulate emotions or apply cognitive tools effectively between scheduled therapeutic sessions.

KEY GOALS

- Multimodal interaction supporting combined text and vision inputs for state analysis.
- Interactive therapeutic workflows built around active Cognitive Behavioral Therapy (CBT) and Dialectical Behavior Therapy (DBT) protocols.
- Reliable safety guardrails to accurately flag high-risk behaviors and perform crisis-level escalation.

APPLICATION AREAS

Digital Therapeutics, Mental Health Applications, Outpatient Support Systems, Wellness Platforms.

PROJECT SUMMARY

A vision-enabled AI companion reinforcing CBT/DBT strategies through contextual chat, text evaluations, and emotional assessments to provide ongoing care infrastructure.

Medication Agent

Supervisor: **Raj Kumar Biswokarma** • Client: **Individual/Person** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Medication non-adherence, incorrect dosages, and dangerous drug-to-drug interactions present severe healthcare risks, particularly for vulnerable seniors managing complex multi-prescription routines.

KEY GOALS

- Visual recognition and verification of physical pills, capsules, and pharmaceutical label instructions.
- Parsing of diverse medical schedules and validation against integrated safety/interaction databases.
- Deployment of a persistent conversational check-in agent to assist patient tracking.

APPLICATION AREAS

Geriatric Care, Home Healthcare, Pharmacy Automation, Patient Adherence Monitoring.

PROJECT SUMMARY

A vision-based autonomous agent that validates real-time prescription guidelines, cross-checks drug-to-drug interactions, and proactively tracks day-to-day adherence.

AI for Senior Citizen Care

Supervisor: **Rakshya Lama Moktan** • Type: **Prototyping**

PROBLEM STATEMENT

Seniors living independently encounter unmonitored baseline hazards (e.g., sudden physical falls, unmapped cognitive decline) in environments lacking continuous, high-cost medical care infrastructure.

KEY GOALS

- Non-intrusive camera video stream monitoring to detect movement anomalies.
- High-context, empathetic, and compassionate conversational dialogue engine.
- Actionable analytical dashboards built for direct access by families and remote attending physicians.

APPLICATION AREAS

Assisted Living Facilities, Remote Patient Monitoring, Eldertech Platforms, Emergency Response Systems.

PROJECT SUMMARY

An integrated AI system pairing real-time vision-based anomaly and fall detection with a highly conversational, empathetic dialogue model to comprehensively safeguard isolated seniors.

2. MULTI-AGENT SYSTEMS

Recruiter (TA-MAS)

Supervisor: **Pawan Sapkota Sharma** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Traditional corporate recruitment pipelines suffer from massive cognitive bottlenecks, cross-platform data fragmentation, and high manual operational friction.

KEY GOALS

- Lead Orchestration Agent for smooth injection into modern Applicant Tracking Systems (ATS).
- Autonomous Sourcing Agent configured to analyze complex network graphs and candidate listings.
- Automated CV screening, fitment scoring, and intelligent engagement sequencing loops.

APPLICATION AREAS

HR Technology (HRTech), Talent Acquisition Platforms, Corporate Recruitment, Staffing Agencies.

PROJECT SUMMARY

An autonomous multi-agent hiring ecosystem that natively ingests dynamic job definitions, calculates deep candidate fit scores, and orchestrates customized programmatic communications.

PR Intern (Public Relation Officer)

Supervisor: **Pawan Sapkota Sharma** • TA: **Akshath B. Chhetri** • Type: **Agentic, Specifications**

PROBLEM STATEMENT

Modern corporate communication is heavily reactive. Delayed, manual responses to rapid digital PR crises can result in widespread, permanent brand health damage.

KEY GOALS

- Always-on media sentiment monitoring across global networks and public pipelines.
- RAG-driven press release draft generation tailored to contextually isolate unfolding corporate threats.
- Semantic journalist matching to ensure structured, highly targeted communication outreach.

APPLICATION AREAS

Public Relations & Communications, Brand Risk Management, Marketing Automation, Media Monitoring Services.

PROJECT SUMMARY

An enterprise monitoring and execution framework that tracks international media feeds, evaluates live brand health, and issues targeted communications under human-in-the-loop controls.

PM OS

Supervisor: **Raj Kumar Biswokarma** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Project management applications suffer from a sharp disconnect between abstract task documentation (e.g., Jira tickets) and the concrete, live technical execution found inside codebase repositories.

KEY GOALS

- Continuous data synchronization across code repositories, version control systems, and communication threads.
- Real-time velocity drift prediction paired with dynamic dependency graph analysis.
- Automated project progress reporting and proactive engineering task drafting.

APPLICATION AREAS

Software Development Lifecycle (SDLC), Enterprise Agile Management, Remote Team Operations, SaaS Tools.

PROJECT SUMMARY

An autonomous operational framework that leverages coordinated vision-and-language agents to dynamically monitor software lifecycles, map execution states, and systematically mitigate delivery risks.

In-Browser AI Agent

Supervisor: **Bimarsha Khanal** • Type: **Prototyping**

PROBLEM STATEMENT

Cloud-dependent web-automation agents introduce high data privacy risks, cause significant network latency overheads, and result in substantial ongoing cloud operational infrastructure costs.

KEY GOALS

- Local execution optimization via WebGPU/WebNN architectures utilizing Small Language Models (SLMs).
- Live DOM element parsing and direct mapping to native client-side web browser execution actions.
- Strict sandbox boundaries to reinforce local user data isolation.

APPLICATION AREAS

Local/Edge AI Computing, Privacy-First Web Browsing, Robotic Process Automation (RPA), Desktop Assistants.

PROJECT SUMMARY

A client-side browser assistant running natively on hardware via local SLM integration, executing multi-step internet actions while keeping data fully private.

Smart Form Auditor

Supervisor: **Pawan Sapkota Sharma** • Type: **Vision, Prototyping**

OPERATIONAL CONTEXT & SCOPE

Tracked in the directory as an advanced vision-centric multi-agent prototype. It focuses on validating visual layouts, processing document entries, and executing structural auditing checks on digitizable data streams.

3. GOVERNANCE & DEVELOPMENT (GOVTECH)

Help Me Navigate

Supervisors: [S. Kisee](#), [R. Lama Moktan](#) • Type: [Vision](#), [Prototyping](#)

PROBLEM STATEMENT

Navigating complex, highly packed public facility layouts (e.g., healthcare compounds, sprawling municipal wings) induces severe corridor congestion, physical delays, and heavy administrative processing overheads.

KEY GOALS

- Computer vision system to parse layout signage, informational boards, and static map directories.
- Localized, hardware-agnostic pathfinding and path generation optimized for mobile devices.
- Real-time dynamic audio/text situational navigation overlays.

APPLICATION AREAS

Smart Cities & Civic Tech, Accessibility & Wayfinding, Hospital Operations, Public Facility Infrastructure.

PROJECT SUMMARY

An internal localization and pathfinding system leveraging edge computer vision processing to smoothly guide citizens through complex municipal and medical physical infrastructures.

Ward Office Assistant

Supervisor: [Shreejan Kisee](#) • Type: [Language](#), [Project Planning](#)

PROBLEM STATEMENT

Local ward-level government management in Nepal is slowed by intricate administrative requirements, fragmented document protocols, and lack of clarity on local rules for public applications.

KEY GOALS

- Development of an expert agent specialized in local government regulations and application criteria.
- Adaptive dynamic checklist compilation based on specific citizen requests or status profiles.
- Localized NLP optimization to streamline citizen support interactions in native dialects.

OPERATIONAL SCOPE

Incorporates functional protocols designed to explicitly identify, map, and query representatives from any given local ward sector.

APPLICATION AREAS

GovTech, Civic Tech, Local Governance Administration, Public Bureaucracy Streamlining.

PROJECT SUMMARY

A localized administrative assistant that processes civic paperwork parameters and guides citizens step-by-step through local municipal documentation requirements.

Fairness & Bias Audit

Supervisor: [Shreejan Kisee](#) • Type: [Vision](#), [Prototyping](#)

PROBLEM STATEMENT

Opaque, unmonitored computer vision classifiers and automated decision frameworks risk quietly scaling structural biases, demographic disparities, and errors against minority populations.

KEY GOALS

- Creation of dense, multi-demographic validation testing matrices for complex image classifiers.
- Statistical detection of latent data skews and embedding distance disparities.
- Automated engineering paths for standard compliance auditing report generation.

APPLICATION AREAS

Responsible AI, AI Ethics, Algorithmic Compliance Auditing, Regulatory Risk Management.

PROJECT SUMMARY

An end-to-end diagnostic environment that interrogates deployed vision systems for hidden bias patterns to satisfy international regulatory compliance requirements.

Public Policy Microsimulation

Supervisor: **Bimarsha Khanal** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Traditional static macroeconomic models fail to capture adaptive human behavioral shifts, secondary feedback loops, and geographic adjustments triggered by legislative or tax restructuring.

KEY GOALS

- Modeling a dense, highly populated multi-agent simulation framework.
- Predicting long-term systemic reactions and fiscal adjustments following policy changes.
- Spatial computer vision tracking to map and evaluate physical socio-economic geographic migrations.

APPLICATION AREAS

Macroeconomic Policy Planning, Behavioral Economics, Regulatory Impact Assessment, Urban & Spatial Planning.

PROJECT SUMMARY

An advanced predictive multi-agent micro-simulation framework that models complex human behaviors to provide empirical insights for sustainable legislative designs.

Nepali Document Authoring & Verification

Supervisor: **Shreejan Kisee** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Manual formulation and verification of massive legal archives and document loads in government offices remain slow, prone to systemic errors, and highly vulnerable to malicious fraud.

KEY GOALS

- Layout-aware automated document drafting engine optimized for local administrative templates.
- Optical character and data structure extraction directly from analog paper scans and digital records.
- Advanced contradiction and document tampering/fraud detection pipelines.

APPLICATION AREAS

LegalTech in Nepal, Notary & Authentication Automation, Judicial Processing, Fraud Prevention.

PROJECT SUMMARY

A secure operational platform combining structural vision models and NLP to assist administrative personnel in producing valid documents and cross-checking records for tampering.

4. SOFTWARE ENGINEERING & RETAIL TECHNOLOGY

Data Engineering Intern

Supervisor: **Pawan Sapkota Sharma** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Data architecture teams waste substantial engineering hours generating boilerplate ETL infrastructure, manually repairing broken schemas, and resolving brittle data pipelines.

KEY GOALS

- Parsing architectural diagrams, database structures, and system schemas to automate ETL pipeline generation.
- Real-time data-drift tracking coupled with auto-patching runtime adjustments.
- Computer vision interpretation of complex visual database health dashboards.

APPLICATION AREAS

Data Operations (DataOps), Pipeline Observability, Developer Productivity Tools, Automated Architecture.

PROJECT SUMMARY

An autonomous assistant that reads complex technical drawings and live monitoring panels to build, optimize, and automatically repair critical backend data pipelines.

Voice Anonymizer

Supervisor: **Bimarsha Khanal** • Type: **Audio, Research**

PROBLEM STATEMENT

The rapid spread of scalable biometric profiling engines and AI deepfakes compromises speech data privacy, exposing confidential audio data and identities to exploitation.

KEY GOALS

- Isolating and removing unique physiological speaker biometric markings from raw wave formats.
- Preserving structural semantic definitions, linguistic clarity, and vocal emotional inflections.
- Executing adversarial acoustic filtering techniques to prevent unauthorized re-identification.

APPLICATION AREAS

Privacy Engineering, Cybersecurity, Whistleblower Support, Biometric Spoofing Countermeasures, Secure Telecommunications.

PROJECT SUMMARY

An audio processing and research tool configured to strip structural identity properties from recordings, maintaining perfect semantic integrity while safeguarding caller privacy.

Same Product Identification

Supervisor: **Pawan Sapkota Sharma** • Type: **Vision, Prototyping**

PROBLEM STATEMENT

Multi-vendor marketplace platforms (e.g., Daraz) face heavy catalog clutter due to duplicate listings, disjointed asset specifications, and untracked retail pricing variances.

KEY GOALS

- Fine-grained multi-perspective visual alignment across varying item display photographs.
- Integrating deep textual similarity architectures to reconcile diverse dealer descriptions.
- Automated catalog data querying to capture, map, and flag pricing variances.

APPLICATION AREAS

E-commerce Catalog Management, Search Engine Optimization, Retail Pricing Intelligence, Cross-Vendor Analysis.

PROJECT SUMMARY

A specialized processing infrastructure blending computer vision and NLP pipelines to cluster identical inventories across e-commerce channels, bringing transparency to marketplace listings.

5. EDUCATION & ENTERPRISE SYSTEMS

Content Authoring Workspace

Supervisor: **R. Lama Muktan** • Partner: **ADB Team & R. Shikhrakar** • Type: **Vision, Research**

PROJECT CONTEXT & COLLABORATION

In direct partnership with the ADB Team and Rojesh Shikhrakar, this research track establishes comprehensive automated publishing frameworks using unified vision-language execution spaces.

PROBLEM STATEMENT

Educators and curriculum designers in Nepal face a critical gap in localized software required to format and build enriched content in the Nepali script. Matching core text blocks with accurate pedagogical graphics requires slow manual editing and graphic design expertise.

KEY GOALS

- Language-aware parsing architecture optimized for Nepali script, grammar rules, and textbook formatting.
- Computer vision model integration to contextually suggest or produce accurate graphical imagery aligned with lesson text.
- Automated validation checking to ensure strict national curriculum compliance and age-appropriate reading difficulty scores.

APPLICATION AREAS

Educational Publishing Houses, Localized EdTech Suites, National Curriculum Systems, Interactive Teacher Tools.

PROJECT SUMMARY

A specialized publishing environment blending Nepali NLP modeling and image analysis to allow rapid authoring of verified, illustrated learning content matching regulatory benchmarks.

Nepali Voice Learning App

Supervisor: **Bimarsha Khanal** • Type: **Audio, Research**

PROBLEM STATEMENT

Standard language platforms lack acoustic clarity and script verification systems calibrated for native Nepali speech patterns, preventing learners from receiving real-time evaluation across reading, writing, listening, and speaking modules.

KEY GOALS

- Training specialized Automated Speech Recognition (ASR) setups tuned to local dialects and phonetics.
- Building highly realistic text-to-speech (TTS) frameworks for natural audio playback.
- Deploying low-latency acoustic models to evaluate pronunciation accuracy and log individual mastery charts.

APPLICATION AREAS

Primary Literacy Infrastructure, Language Preservation Projects, Distance Learning Suites.

PROJECT SUMMARY

An interactive, multi-modal literacy engine providing speech and script analysis to guide native and non-native learners through advanced pronunciation modules.

AI Accessibility System (Subjective Grader for Braille)

Supervisor: **Raj Kumar Biswokarma** • Type: **Prototyping**

OPERATIONAL CONTEXT & SCOPE

An accessibility project focusing on computer vision and document parsing to read, translate, and evaluate Braille documentation, providing an automated grading interface for subjective assessments.

Enterprise InfoSec Multi-Agent Framework

Category: **Advanced Governance** • Type: **Agentic Track**

SYSTEM DEPLOYMENT SETTING

Configured to operate directly within enterprise SOAR (Security Orchestration, Automation, and Response) and GRC (Governance, Risk, and Compliance) systems as an autonomous auditing and infrastructure engine.

PROBLEM STATEMENT

Fragmented security log data causes slow corporate compliance tracking and delays vulnerability isolation. This floods Security Operation Centers (SOCs) with manual triage burdens and leaves critical vulnerability windows unaddressed.

INTEGRATED SUB-AGENT ARCHITECTURE

- **Central Orchestrator Agent:** Evaluates overall infrastructure state parameters, routes anomalies, and directs specialized sub-agents.
- **Compliance Sub-Agent:** Performs continuous checks of runtime systems against global regulations (ISO 27001, SOC2, NIST).
- **Vulnerability Triaging Sub-Agent:** Queries live CVE data channels to calculate accurate, context-specific corporate risk metrics.
- **Incident Orchestration Sub-Agent:** Automatically triggers isolation and remediation routines under human validation rules.

APPLICATION AREAS

Cloud Security Posture Management (CSPM), Automated Auditing, Vulnerability Lifecycles, Enterprise GRC Reporting.

Continuous Compliance & Audit Automation Engine

Category: **Enterprise Architecture** • Type: **Multi-Agent Execution**

PROBLEM STATEMENT

Aligning internal technical execution with evolving global regulations remains a manual, reactive, and error-prone process. Because compliance markers are buried deep within low-level log states and code structures, teams fail to detect compliance drift in real-time.

COORDINATED AGENT REGISTRY

- **Compliance Orchestrator Lead:** Monitors internal network systems, alerts human endpoints of live violations, and structures final report outputs.
- **Regulatory Ingestion & Delta Agent:** Scrapes, structures, and logs modifications to active legal frameworks and statutory rules.
- **Audit Scanner Agent:** Executes continuous, deterministic scans across corporate log pools, codebases, and cloud architectures.
- **Policy Evaluator Agent:** Maps infrastructure states against current legal constraints to auto-generate audit-ready reporting sheets.

PROJECT SUMMARY

An interconnected multi-agent framework designed to automate enterprise governance by cross-referencing global rule adjustments against internal execution metrics to systematically prevent compliance violations.